

ABHISHTA H MALLAYA

+91-8072819574 | abhishtamallaya@gmail.com | linkedin.com/in/abhishta-h-mallaya | github.com/Abhishta-H-Mallaya | Coimbatore, India

PROFESSIONAL SUMMARY

Final-year B.Tech student in Artificial Intelligence & Data Science (CGPA 8.56/10) with hands-on experience building end-to-end AI pipelines from agentic LLM orchestration and multimodal embedding-based systems to computer vision and quantum machine learning. Comfortable across the data lifecycle with Python, SQL, and BI tooling (Power BI, Tableau), and experienced integrating model inference with real-time, sensor-to-alert deployment. IEEE SISCON 2025 Best Paper awardee, with a track record of turning research prototypes into working, deployed systems.

EDUCATION

Amrita Vishwa Vidyapeetham

Coimbatore, Tamil Nadu

B.Tech in Artificial Intelligence & Data Science | CGPA: 8.56 / 10

Aug 2023 – Jul 2027

TECHNICAL SKILLS

- Languages & Query: Python, SQL (PostgreSQL), HTML, CSS
- AI / LLM & Agentic Systems: LangGraph (multi-agent orchestration), Qwen2.5-7B, HuBERT & SpeechT5 embeddings, TensorFlow, PyTorch
- Machine Learning / Computer Vision: CNNs, LSTMs, ArcFace, CBAM Attention, ResNet-50, VGG-16, Vision Transformers, Hybrid Quantum-Classical Models
- Quantum Computing: Qiskit, Variational Quantum Circuits, Ansatz Optimization
- Data & BI: Feature Engineering, Power BI, Tableau, Grad-CAM, GLCM
- Tools: Git/GitHub, VS Code, Raspberry Pi, Arduino

EXPERIENCE

QMET Tech Foundation – PARIMANA Summer Internship

Remote

Research Intern, Quantum Sensing, Imaging & Metrology

Summer 2026

- Competitively selected for a research program in Quantum Sensing, Imaging & Metrology; working on real-world quantum technology applications in collaboration with domain researchers.
- Developed a Python-based preprocessing pipeline for wearable OPM-MEG data, enabling reproducible signal and sensor characterization.
- Implemented statistical and spectral analysis techniques for empirical noise floor estimation and signal quality assessment.
- Benchmarked sensor performance against theoretical quantum sensitivity limits through computational modeling and literature-driven analysis.

Institution's Innovation Council – Amrita Vishwa Vidyapeetham

Coimbatore

Innovation Ambassador & R&D Cell Member

Jan 2026 – Present

- Trained in IPR, startup ecosystems, and research ethics; coordinated mentor-mentee programs, academic events, and contributed to active R&D projects through development and technical reviews.

Anokha Tech Fest – Amrita Vishwa Vidyapeetham

Coimbatore

Co-Head, TechFair & Exhibitions

Nov 2025 – Jan 2026

- Led logistics planning for institution-wide student participation and coordinated large-scale technical exhibitions.

PUBLICATIONS

Sentinel Turret Rover: Intruder Defense System

Apr 2025

Python, ArcFace, CBAM-CNN, OpenCV, Raspberry Pi

- Best Paper Award – IEEE SISCON 2025 (IoT in Smart Systems track).
- Designed an autonomous surveillance rover turret using ArcFace face recognition integrated with a CBAM-enhanced CNN, achieving 95% identification accuracy.
- Deployed an optimized real-time inference pipeline on Raspberry Pi for low-latency surveillance with full sensor-to-alert output integration via Telegram Bot.

KEY PROJECTS

Agentic LLM-Based SLP System for Neurological Speech Disorder Diagnosis & Voice Restoration

Mar 2026

LangGraph, HuBERT, SpeechT5, TensorFlow, Qwen2.5-7B

- Developed a three-module agentic AI system for neurological speech disorder diagnosis, personalized speech-language therapy planning, and voice restoration via voice cloning.
- Integrated HuBERT speech embeddings, LLM-based reasoning, SpeechT5 voice synthesis, and multimodal clinical analysis, achieving 88.9% diagnostic accuracy for ALS, Parkinson's disease, and Dysarthria.

Quantum Machine Learning for Drug Screening

Sep 2025

Hybrid Quantum Neural Networks, Variational Quantum Algorithms, Qiskit, PyTorch

- Designed a hybrid quantum-classical deep learning framework for phenotypic drug screening on the BBBC021 biomedical imaging dataset, integrating variational quantum circuits with neural networks.
- Achieved 93.75% classification accuracy by optimizing a 48-qubit ansatz, demonstrating the effectiveness of quantum-enhanced learning for pharmaceutical image analysis.

Herbal Leaf Classification using ResNet-50 and UAV Imaging

Nov 2024

ResNet-50, Vision Transformers, Deep Learning, GLCM, Computer Vision

- Curated a 6,904-image dataset spanning 77 medicinal plant species and built a deep learning framework for automated classification from UAV-acquired aerial imagery.

- Achieved 97.68% accuracy using ResNet-50 with transfer learning and GLCM-based feature engineering, optimizing the model for real-time inference in precision-agriculture applications; outperformed a VGG-16 baseline (47.4%) by over 50 points.

CERTIFICATIONS & ACHIEVEMENTS

- Certified Blockchain Associate (CBA) – Kerala Blockchain Academy (Jun 2026): blockchain fundamentals, decentralized systems, smart contracts, and consensus mechanisms.